

XENIOS MVP

Product Requirements Document

1. Executive Summary

1.1 What We Are Building

Xenios is the first AI coaching intelligence platform that transforms how personal trainers and health coaches operate. In six weeks, we will deliver a production-ready MVP that combines automated data extraction from health documents, real-time session recording with AI-powered transcription, and intelligent coaching analytics—eliminating 10+ hours of administrative work per week while making trainers measurably better at their craft.

The platform integrates three core capabilities into a unified system: (1) an intelligent data extraction engine that automatically processes InBody scans, lab results, wearable exports, and nutrition logs into structured, actionable insights; (2) meeting intelligence that captures in-person training sessions via audio recording, transcribes conversations, and generates comprehensive session summaries in under 90 seconds; and (3) coaching performance analytics that provide trainers with objective feedback on their communication patterns, cueing effectiveness, and client engagement trends.

1.2 Who It Is For

Primary Users: Elite personal trainers and health coaches managing 20-200+ clients who want to scale their practice without sacrificing service quality. These professionals currently spend 15-20 minutes after each session manually documenting workouts, form cues, and client conversations—time that could be spent training additional clients or improving their craft.

Secondary Users: Clients of these coaches, who receive personalized session recaps, progress tracking, and a more connected coaching experience. The platform enhances the coach-client relationship by ensuring nothing discussed in sessions is forgotten and every milestone is celebrated.

1.3 Core Problems Solved

- **Documentation Burden:** Trainers spend 15-20 minutes per session on manual note-taking. With Xenios, this drops to under 2 minutes of review time—a 90% reduction that saves 5-10 hours weekly for full-time trainers.
- **Fragmented Health Data:** Client information lives across multiple apps, PDFs, and spreadsheets. Xenios consolidates everything into a single source of truth with AI-generated insights.
- **No Coaching Feedback Loop:** Trainers receive zero systematic feedback on their coaching effectiveness. Xenios provides objective analytics on communication patterns, cueing frequency, and client sentiment trends.
- **Client Churn Blindspots:** Coaches often don't realize a client is disengaging until they've already left. Xenios detects early warning signals and prompts proactive intervention.

1.4 Success Metrics

The MVP will be considered successful when it achieves the following criteria:

Metric	Target
Time saved per session	13-18 minutes (vs. 15-20 min manual)
Session summary generation time	< 90 seconds end-to-end
Workout extraction accuracy	90%+ on exercises, sets, reps, weights
Transcription accuracy (gym environment)	≥ 85% with quality headset mic
Coach satisfaction (NPS)	≥ 50 among pilot coaches
System uptime	99.5%+ availability

The "Wow" Moment: *A trainer finishes a 45-minute session, taps 'Complete,' and within 60 seconds receives a fully structured summary with exercises, form cues, consultation notes, and a draft message to send their client. What previously took 15-20 minutes of tedious typing is done before they can refill their water bottle.*

2. Real-World Use Cases

The following scenarios demonstrate how Xenios solves real problems for personal trainers in their daily workflow.

2.1 Use Case: Complete In-Person Training Session

Context: Sarah is a 34-year-old client working with Coach Mike. She's 12 sessions into a 16-week program with a goal to lose 15 lbs before her wedding in 8 weeks. Last session, she mentioned lower back tightness during deadlifts.

Pre-Session (Automated Preparation)

Mike arrives at the gym and opens the Xenios mobile app. He immediately sees Sarah's session scheduled with a quick prep brief that includes: her last session notes (lower back tightness during deadlifts), current progress (squat increased to 125 lbs, ready for 135?), goal status (9 lbs lost, 6 lbs remaining in 3 weeks), and personal context (son has soccer tournament this weekend).

Pain Point Addressed: Without Xenios, Mike would need to scroll through text messages, check separate apps for workout history, and try to remember personal details about Sarah's life—often forgetting important context that builds client relationships.

During Session (Seamless Capture)

Mike taps 'Start Session' and puts his phone in his pocket. He wears a fitness-grade Bluetooth headset mic (recommended: Samson Airline 77 or similar). For the next 45 minutes, he coaches Sarah exactly as he normally would—no behavior change required. The app quietly records the session audio in the background.

Pain Point Addressed: Current solutions require trainers to stop mid-session to log exercises, breaking the coaching flow. Xenios captures everything passively—the trainer's full attention stays on the client.

Post-Session (The Magic)

Mike taps 'Complete Session.' Within 60 seconds, a structured summary appears on his phone containing: workout completed (exercises with exact weights recorded), form cues extracted ('chest up' mentioned 4 times during squats), consultation notes (wedding stress, reassurance provided, sleep improving to 7 hours), personal connection reminders (follow up about son's tournament on Monday), and a draft personalized message ready to send to Sarah.

Mike reviews the summary in 45 seconds, makes one small edit (adds a fire emoji to the message), and taps 'Send to Sarah.' Total post-session time: under 2 minutes.

2.2 Use Case: Coaching Performance Analytics

Context: Mike has been using Xenios for 12 weeks. He has 35 active clients and has conducted 420 sessions through the platform.

Mike logs into the web dashboard and navigates to 'My Coaching Analytics.' He discovers objective data about his coaching style that he never had access to before: his talk-time ratio is 58% coach / 42% client (benchmark: elite trainers average 55/45); his most common form cue is 'chest up' (mentioned 347 times this month); his clients' positive sentiment is 84% (up 6% from last month); his average message response time is 6.2 hours (top performers: 3.8 hours).

2.3 Use Case: Risk Detection Prevents Client Churn

Context: Mike has 35 clients. One is about to ghost. He doesn't know it yet.

Mike opens the Xenios app Tuesday morning and sees a red notification badge: '3 clients showing disengagement patterns.' He clicks into the Client Risk Dashboard and sees James Rodriguez flagged with an 87/100 risk score (High Risk).

The system explains why: session completion dropped from 100% to 75% over the past two weeks; message response time increased from 2 hours to 18 hours; sentiment analysis shows positive language down 42%; and James mentioned being 'busy' three times in recent sessions.

2.4 Use Case: Data Extraction and Insights

Context: A new client, Marcus, just completed his onboarding assessment. He's uploaded an InBody body composition scan, recent bloodwork from his doctor, and connected his WHOOP wearable.

Within 60 seconds of upload, Xenios automatically extracts structured data from all sources: from the InBody PDF, it pulls weight (185.4 lbs), body fat percentage (22.3%), skeletal muscle mass (78.2 lbs), and basal metabolic rate (1,847 kcal). From the lab results CSV, it extracts metabolic markers (fasting glucose: 98 mg/dL, flagged as borderline high), lipid panel (LDL: 142 mg/dL, flagged as elevated), and testosterone levels. From WHOOP, it aggregates 7/14/30-day averages for HRV, sleep duration, recovery scores, and strain levels.

3. Technical Architecture

The Xenios MVP follows a monolithic architecture optimized for a small engineering team building in 6 weeks. This approach eliminates operational complexity while maintaining clear scaling paths for post-MVP growth.

3.1 System Overview

The system consists of five primary layers:

- **Mobile Application (React Native):** Cross-platform app for session recording, client management, and real-time notifications. Handles Bluetooth headset mic integration, background audio recording, and offline-first data sync.
- **Web Application (Next.js):** Dashboard for analytics, detailed session review, and administrative functions. Provides rich data visualization and bulk operations not suitable for mobile.
- **Backend API (Node.js/Express or Next.js API Routes):** Unified API layer handling authentication, business logic, and orchestration of AI services. Includes RBAC enforcement and comprehensive audit logging.
- **Background Workers (pg-boss):** PostgreSQL-native job queue for async processing: transcription jobs, document extraction, insight generation, analytics aggregation, and risk detection.
- **External Services:** Deepgram for ASR/transcription, IBM Docling for document extraction, Claude/GPT-4 for NLP and summarization, S3/R2 for file storage.

3.2 Component Communication

All client applications communicate with the backend via HTTPS REST APIs. Real-time updates (transcription streaming, notification delivery) use WebSocket connections. The backend orchestrates external service calls and manages job queues.

Authentication Flow: JWT-based authentication with refresh tokens. Mobile app stores tokens securely in device keychain. All API requests include authorization headers validated against PostgreSQL-backed session store.

3.3 External Services and Integrations

Service	Provider	Purpose
Speech-to-Text	Deepgram Nova-2	Real-time transcription with speaker diarization
Document Extraction	IBM Docling (self-hosted)	PDF parsing for InBody scans, lab results
LLM Inference	Claude 3.5 Sonnet / GPT-4o	Summarization, classification, entity extraction
File Storage	AWS S3 or Cloudflare R2	Audio files, documents, exports
Database	PostgreSQL (Supabase/Neon)	Primary data store with RLS
Job Queue	pg-boss	Async job processing (PostgreSQL-native)
Push Notifications	Firebase Cloud Messaging	Mobile push notifications

3.4 Technology Stack Decisions

Document Extraction: IBM Docling (Recommended)

After evaluating Azure Document Intelligence, AWS Textract, Google Document AI, and IBM Docling, we recommend IBM Docling as the primary document extraction engine for MVP.

Technology	Cost/1K Pages	Table Accuracy	Recommendation
IBM Docling	\$0 (self-hosted)	97% TEDS	PRIMARY - Best value
Azure Doc Intelligence	\$30	~95%	BACKUP - If Docling fails
AWS Textract	\$15-70	90-95%	Not recommended
Google Document AI	\$30-65	~94%	Not recommended

Transcription: Deepgram Nova-2 (Recommended)

For gym environment recording with speaker diarization, Deepgram Nova-2 provides the optimal combination of streaming latency (<300ms), free speaker diarization, noise robustness, and HIPAA compliance.

3.5 Major Technical Constraints

- Audio Quality Dependency: Transcription accuracy is highly dependent on microphone quality and placement. Phone microphones in gym environments yield 60-70% accuracy; fitness headset mics yield 85-90%.
- Processing Latency: The 60-90 second target for session summary generation requires efficient pipeline orchestration.
- Mobile Background Processing: iOS and Android impose strict limits on background audio recording.
- Data Sovereignty: All PHI must be processed by HIPAA-compliant vendors with signed BAAs.

4. Enhanced Data Model

The data model is designed around three core principles: (1) normalized structure for query performance and data integrity; (2) comprehensive audit logging for compliance; and (3) flexible JSONB fields only where schema evolution is genuinely needed.

4.1 Core Entities

Users, Coaches, and Clients

The system uses a unified users table with role-based differentiation. A user can be a coach, a client, or both.

Entity	Key Fields	Purpose
users	id, email, role, created_at	All system users (coaches and clients)
coach_profiles	user_id, business_name, certifications	Extended profile data for coaches

client_profiles	user_id, goals, program_start_date	Extended profile data for clients
coach_client_relationships	coach_id, client_id, status, started_at	Links coaches to their clients

Sessions and Transcripts

Entity	Key Fields	Purpose
sessions	coach_id, client_id, started_at, processing_status, summary (JSONB)	Training session metadata and generated summary
transcript_segments	session_id, speaker, start_ms, end_ms, text, intent, confidence	Individual utterances with AI classification
workout_exercises	session_id, exercise_name, sets, reps, weight, form_cues_given	Extracted workout data from sessions
form_cues_tracking	session_id, coach_id, client_id, cue_text, exercise_name, cue_count	Coaching cue analytics aggregation

4.2 Entity Relationships

- Users → Sessions: A coach has many sessions; a client has many sessions. Sessions link both via foreign keys.
- Sessions → Transcript Segments: One session has many transcript segments (one per utterance). Cascade delete on session removal.
- Sessions → Workout Exercises: One session has many exercises. Each exercise can link back to specific transcript_segment_ids.
- Clients → Measurements: One client has many measurements. Measurements optionally link to the artifact they were extracted from.
- Insight Cards → Approvals: Each insight card has approval state. Only approved insights are visible to clients.

4.3 Data Rules and Constraints

- Row-Level Security (RLS): Coaches can only access their own clients' data. RLS policies enforce this at the database level.
- Soft Deletes: User accounts and sessions use soft deletion (deleted_at timestamp) rather than hard deletion.
- Audio Retention: Audio files are automatically deleted 24 hours after transcription completion.
- Insight Approval Gate: AI-generated insight cards have status = 'draft' by default. They MUST be approved by a coach before becoming visible to clients.
- Audit Log Immutability: The events_audit table is append-only. No UPDATE or DELETE operations are permitted.

5. 6-Week Development Timeline

The following timeline assumes a team of 2-3 engineers working full-time. The CTO will assign specific responsibilities based on team strengths.

Week 1: Foundation and Infrastructure

Goal: Core infrastructure operational, basic authentication working, database schema deployed.

1. PostgreSQL database setup with Supabase or Neon (all tables created, RLS policies active)
2. Authentication system with JWT tokens and refresh token rotation
3. CI/CD pipeline configured (GitHub Actions → staging environment)
4. pg-boss job queue infrastructure operational
5. S3/R2 storage configured with presigned URL generation
6. Mobile app skeleton (React Native): auth flow, navigation structure
7. Web dashboard skeleton (Next.js): login, basic layout

Week 1 Demo: User can log into mobile app and web dashboard. Database queries return test data. File upload to S3 works.

Week 2: Data Extraction Pipeline

Goal: Upload health documents → automatic extraction → data visible in UI.

8. IBM Docling integration for PDF extraction (InBody scans, lab results)
9. CSV/JSON parsers for wearable exports (Garmin, WHOOP, Apple Health)
10. Async extraction job workers (artifact upload → trigger extraction)
11. Client profile page showing extracted measurements
12. Basic insight generation (flag out-of-range lab values)
13. Coach approval queue UI (draft insights list with approve/reject buttons)

Week 3: Audio Recording and Transcription

Goal: Record sessions on mobile → upload → receive transcription.

14. Mobile session recorder with Bluetooth headset mic support
15. Background audio recording (iOS AVAudioSession, Android Foreground Service)
16. Audio upload flow with progress indicator and offline queuing
17. Deepgram integration for batch transcription with speaker diarization
18. Transcript segment storage (speaker, timestamps, text, confidence)
19. Basic transcript viewer in web dashboard

Week 4: AI Classification and Summarization

Goal: Transcripts → classified segments → structured session summaries.

20. Intent classification pipeline (workout_instruction, form_cue, consultation, off_topic)
21. Entity extraction for workout data (exercise names, sets, reps, weights)
22. Form cue extraction and counting
23. Session summary generation (LLM-powered, structured output)
24. Summary review and editing interface
25. Pre-session brief screen (last session context, suggested focus)

Week 5: Client Experience and Analytics

Goal: Client app receives summaries; coach sees analytics dashboard.

- 26. Client app session recap view with push notifications
- 27. Send-to-client workflow (coach approves → client notified)
- 28. Coaching analytics aggregation jobs (talk time, cue patterns, sentiment)
- 29. Coaching analytics dashboard with visualizations
- 30. Client sentiment tracking from consultation segments
- 31. Basic risk detection algorithm (session completion, response time, sentiment trends)

Week 6: Polish, Risk Detection, and Launch Prep

Goal: Production-ready MVP with all features working end-to-end.

- 32. Client risk dashboard with flags and recommended actions
- 33. Audio cleanup job (24-hour retention policy enforcement)
- 34. Error handling, loading states, and offline mode polish
- 35. Performance optimization (database indexing, query tuning, caching)
- 36. HIPAA compliance audit (encryption verification, audit log review)
- 37. Production deployment with monitoring (Sentry, performance tracking)
- 38. Pilot coach onboarding (5-10 coaches for initial feedback)

6. Technical Specifications

6.1 Audio Recording Requirements

Recording Settings

- Sample Rate: 16 kHz minimum (optimal for speech-to-text)
- Bit Depth: 16-bit
- Format: AAC for transmission (smaller files), WAV/PCM for processing
- Channels: Mono (single mic)
- Expected File Size: ~40MB for 45-minute session (AAC compressed)

Hardware Requirements

Microphone	Price	Battery	Notes
Samson Airline 77 Fitness	~\$200	8+ hours	Best value for trainers
SYNCO G2 Max Wireless	~\$150	10+ hours	Budget-friendly option
Shure GLXD14+/SM31	~\$800	12+ hours	Professional grade

6.2 Transcription Pipeline

- Provider: Deepgram Nova-2 (batch mode for uploaded recordings)
- Speaker Diarization: Enabled (2 speakers: coach and client)
- Processing Time: ~1/10th of audio length (45-min session → ~4.5 min)
- Target Accuracy: ≥85% with quality headset mic, ≥80% acceptable
- Retry Policy: 3 attempts with exponential backoff (60s, 120s, 240s)

6.3 AI Classification and NLP

Intent Classification

Each transcript segment is classified into one of four intents using few-shot prompting with Claude 3.5 Sonnet or GPT-4o:

Intent	Description and Examples
workout_instruction	Exercise commands with sets/ reps/ weights. 'Let's do 3 sets of 10 squats at 135.'
form_cue	Technique corrections. 'Chest up. Drive through your heels. Knees out.'
consultation	Goals, feelings, lifestyle discussion. 'How's your sleep been? Any stress this week?'
off_topic	Irrelevant chatter, greetings, gym noise. 'Hey, what's up. Nice weather today.'

6.4 Performance Requirements

Operation	Target	Maximum
Audio upload (45-min session, LTE)	30-60 seconds	120 seconds
Transcription processing	4-6 minutes	10 minutes
Classification + extraction	30-45 seconds	60 seconds
Summary generation	10-15 seconds	30 seconds
Total pipeline (end-to-end)	60-90 seconds	180 seconds
Dashboard page load	<500ms	1000ms
API response (simple queries)	<100ms	200ms

6.5 Security and Compliance

- Encryption at Rest: AES-256 for all stored data (database, file storage)
- Encryption in Transit: TLS 1.2+ for all API communications
- Authentication: JWT with short expiry (15 min), refresh tokens with rotation
- Authorization: Row-Level Security (RLS) in PostgreSQL; application-level RBAC
- Audit Logging: All PHI access logged with user ID, timestamp, action, and entity
- Data Retention: Audio files deleted after 24 hours; transcripts retained indefinitely
- Vendor BAAs: Required from all vendors handling PHI (Deepgram, cloud storage)

6.6 Error Handling

- Transcription Failure: Retry 3x with exponential backoff. After failure, mark session for manual review and notify coach.
- Low Confidence Classification: Segments with <70% confidence are flagged with ⚠️ in the summary for coach review.
- Upload Failure: Audio cached locally on device. Automatic retry when connection restored. User notified of pending uploads.
- LLM Timeout: 30-second timeout per LLM call. On timeout, use simplified summary template without AI elaboration.

7. Acceptance Tests (Demo-Ready Checklist)

The following tests must all pass for MVP Completion. Each test is binary — it either passes or fails.

Core Workflow Tests:

#	Test	Pass Criteria
1	OpenRouter abstraction	Model swap via env variable, no code changes required
2	InBody extraction	Weight and body-fat % stored with units (or N/A with reason)
3	Nutrition CSV	Protein and kcal averages appear in UI and LLM context
4	Chat grounding	Answer cites at least one numerical extracted value
5	Approval gate	Drafts invisible until approved; audit logs editor + timestamp
6	Async re-analysis	New artifact creates draft InsightCard within 60 seconds

AI Notetaker Tests:

#	Test	Pass Criteria
7	Streaming transcript	Live segments persist with accurate timestamps
8	Live tip trigger	Test phrase* creates LiveTip linked to transcript span
9	Tip to insight	Save tip creates draft InsightCard with evidence refs
10	Fallback summary	Uploaded recording creates summary + draft insight

*Test Phrase Clarification: The 'test phrase' is a QA phrase used during acceptance testing — not a wake word like 'Hey Siri.' During testing, the tester speaks a predetermined phrase (e.g., 'Note: client mentioned knee pain during squats') and verifies the system: (1) detects it as a LiveTip trigger, (2) links it to that transcript segment, and (3) allows saving as a draft InsightCard.

Labs & Wearables Tests:

#	Test	Pass Criteria
11	Lab CSV import	Markers populate with values, units, flags in Labs panel
12	Out-of-range lab insight	Above-range LDL-C produces draft InsightCard
13	Wearable CSV import	Rows normalize into Measurements with units/source
14	Wearable aggregations	7/30-day averages compute and display correctly

Security & Final Tests:

#	Test	Pass Criteria
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15	Audit completeness	Upload, approve, publish events logged with user/role/time
16	End-to-end demo	Full scripted flow executes without errors

8. Risk Mitigation Strategies

8.1 Technical Risks

Risk 1: Audio Quality in Noisy Gyms

- Likelihood: High — gym environments have 80-95 dB background music and equipment noise
- Impact: High — poor transcription makes summaries unreliable
- Prevention: Require fitness-grade Bluetooth headset mics; test in real gym environments during Week 3
- Fallback: If accuracy <70%, mark session for manual review; coach can edit summary extensively

Risk 2: Classification Model Inaccuracy

- Likelihood: Medium — fitness conversations have domain-specific vocabulary
- Impact: Medium — misclassification puts workout data in consultation section (or vice versa)
- Prevention: Use few-shot prompting with 20+ carefully curated examples
- Fallback: Human-in-loop design — coach always reviews before sending

Risk 3: LLM Hallucinations

- Likelihood: Medium — LLMs can invent plausible-sounding details
- Impact: High — invented exercises or advice could harm client trust
- Prevention: Strict prompts ('Do not invent details not in transcript'); use structured output schemas
- Fallback: Coach approval gate — nothing reaches clients without explicit approval

8.2 Product Risks

Risk 5: Coaches Reject Recording Workflow

- Likelihood: Medium — some trainers uncomfortable wearing headsets or recording clients
- Impact: High — core value proposition depends on session recording
- Prevention: Emphasize privacy (audio deleted in 24 hours); clear client consent flow
- Fallback: Recording is optional feature — coaches can still use data extraction and manual notes

8.3 Business Risks

Risk 7: HIPAA Applicability Uncertainty

- Likelihood: Medium — personal trainers exist in regulatory gray area
- Impact: High — non-compliance could create legal exposure
- Prevention: Consult healthcare attorney in Week 1 to determine applicability; build HIPAA-ready regardless
- Fallback: If HIPAA applies, we're already compliant; if not, security posture becomes competitive advantage

9. Definition of Done (MVP Completion)

The MVP is complete when ALL of the following conditions are met.

9.1 Feature Completeness

All core features must be functional and tested:

- 39. Data Extraction: InBody PDFs, lab CSVs, and wearable exports correctly extracted and displayed
- 40. Insight Generation: AI-generated insights with coach approval workflow operational
- 41. Session Recording: 45-minute sessions recorded without crashes, with Bluetooth headset support
- 42. Transcription: Audio transcribed with speaker diarization at $\geq 80\%$ accuracy
- 43. AI Classification: Transcript segments classified into workout, cue, consultation, off-topic
- 44. Summary Generation: Structured summaries generated within 90 seconds of session end
- 45. Client Delivery: Summaries delivered to clients via push notification
- 46. Coaching Analytics: Dashboard displaying talk time, cue patterns, sentiment trends
- 47. Risk Detection: At-risk clients flagged with recommended actions

9.2 Quality Requirements

- No Critical Bugs: Zero crashes, data loss, or security vulnerabilities in production
- Performance Targets Met: All operations within specified latency targets
- Pilot Validation: 5+ pilot coaches have successfully recorded real sessions and approved summaries
- Acceptance Tests Passing: All 16 acceptance tests from Section 7 pass
- End-to-End Demo: Complete demo flow runs without errors for external audience

9.3 Non-Functional Requirements

- Uptime: 99.5%+ availability during pilot period
- Security: Encryption at rest and in transit; RLS enforced; audit logging active
- Compliance: HIPAA audit checklist completed; BAAs signed with all PHI-handling vendors
- Monitoring: Error tracking (Sentry) and performance monitoring deployed
- Documentation: API documentation, database schema, deployment guide, and coach onboarding materials complete

9.4 Stakeholder Sign-Off

The following individuals must approve MVP completion:

- CEO: Confirms product meets strategic vision and is ready for public launch
- CTO: Confirms technical architecture is sound and scalable
- Pilot Coaches: At least 3 pilot coaches provide written testimonials confirming value

9.5 Launch Readiness

- Production Environment: Live and stable with automated deployments
- Pricing & Billing: Stripe integration configured for subscription payments
- Support Channel: Dedicated support mechanism operational (Slack, email, or in-app)
- Legal Documents: Terms of Service, Privacy Policy, and consent forms reviewed by legal counsel
- App Store Submission: Mobile app submitted to Apple App Store and Google Play Store (if applicable)

17. Programming Orchestration Layer (Fitness + Health Coaching)

Purpose: Turn extracted data + insights into an actionable plan that can be assigned, progressed, and adapted. Without programming, Xenios risks becoming a very intelligent dashboard. Coaches do not pay for dashboards—they pay for outcomes delivered through a platform that helps them deliver a plan.

The Programming Orchestration Layer is the critical bridge between Xenios's data intelligence capabilities and actual coaching outcomes. It enables coaches to translate AI-generated insights into structured, progressable fitness and behavior programs that clients can follow and that the system can track, adapt, and optimize over time.

17A. Fitness Programming (Sets/Reps/Phases)

The Fitness Programming module provides coaches with tools to build, assign, progress, and track structured workout programs for their clients.

Program Builder

Coaches can create programs using templates (pre-built programs for common goals like fat loss, muscle gain, athletic performance) or freeform design (custom program creation from scratch). The builder supports drag-and-drop exercise organization, exercise library with video demonstrations, auto-suggestions based on client history and goals, and duplicate/clone functionality for similar clients.

Program Hierarchy

Programs follow a structured hierarchy that mirrors how coaches conceptualize training:

Level	Description	Example
Program	The overarching training plan	12-Week Strength Foundation
Phase	A training block with specific focus	Phase 1: Base Building (Weeks 1-4)
Microcycle	Typically one week of training	Week 2: Volume Accumulation
Session	A single workout	Day 1: Lower Body Push
Exercise	Individual movement	Back Squat
Sets	Execution parameters	4 sets × 8 reps @ 185 lbs

Versioning

Program changes are tracked over time with full audit trail:

- Who changed what and why (e.g., 'Reduced squat volume due to knee discomfort - Coach Mike, Jan 15')
- Ability to revert to previous versions if needed
- Diff view showing changes between versions
- Automatic version creation on significant modifications

Client-Facing Delivery

Clients receive their programs through:

- Weekly schedule view showing upcoming sessions
- Individual session view with exercise details, videos, and notes
- Progress indicators showing where they are in the program
- Quick-start button to begin today's session

Adherence Tracking

For each session and exercise, the system tracks:

Metric	Description	Data Type
Completion Status	Completed, Skipped, Modified, Partial	Enum
Actual Performance	Sets, reps, weight actually performed	Numeric
RPE (Rate of Perceived Exertion)	Client's subjective difficulty rating (1-10)	Scale
Notes	Free-text comments from client	Text
Pain Flags	Did client experience pain? Location and severity	Structured + Text
Modification Reason	Why was exercise modified or skipped?	Categorical + Text

17B. Behavior Programming (Weekly/Monthly Cues and Goals)

Beyond workout programming, sustainable results require behavior change. The Behavior Programming module helps coaches assign, track, and adapt lifestyle interventions.

Goal Builder

Coaches can set one to three behavior targets per week for each client. The system enforces a 'minimal effective dose' philosophy—too many goals leads to overwhelm and failure.

- Goal categories: Sleep, Nutrition, Hydration, Movement, Stress Management, Recovery
- SMART goal framework: Specific, Measurable, Achievable, Relevant, Time-bound
- Progressive difficulty: Goals can be escalated as habits solidify
- Client input: Clients can propose goals for coach approval

Cue Library

A database of tiny actions and environment prompts that support behavior change:

- Tiny actions: 'Take 3 deep breaths before meals' or 'Set phone to Do Not Disturb at 9pm'
- Environment prompts: 'Put gym bag by the door' or 'Keep water bottle on desk'
- Reminder scheduling: Configurable push notifications at optimal times
- Cue effectiveness tracking: Which cues lead to goal completion?

Behavior Tracking

Metric	Description	Purpose
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Daily Check-in	Yes/No completion for each goal	Simple accountability
Streak Tracking	Consecutive days of completion	Motivation through momentum
Friction Notes	What made it hard today?	Identify barriers
Success Notes	What made it easy?	Identify enablers
Weekly Summary	Completion rate + patterns	Coach review dashboard

Escalation Rules

When adherence drops below thresholds, the system suggests simplifying:

- If completion rate < 70% for 2 weeks → System drafts 'simplification suggestion' for coach review
- Suggested simplifications: Reduce number of goals, reduce goal difficulty, change cue timing
- Coach approves/modifies suggestions before client sees them (maintains approval gate pattern)

17C. AI-Assisted Updates (The Loop)

This is where Xenios's intelligence truly shines—automatically suggesting program adjustments based on the comprehensive data the system collects.

Trigger Sources

Program adjustment drafts are triggered by:

- Intake data: New InBody scan, lab results, or wearable data showing meaningful change
- Check-ins: Weekly or daily client check-in responses indicating issues
- Session data: Transcripts revealing pain, fatigue, or motivation concerns
- Messages: Client messages mentioning relevant factors (travel, illness, stress)
- Wearable trends: Declining HRV, poor sleep patterns, elevated resting HR
- Adherence patterns: Consistently missed sessions or modified exercises

Program Adjustment Draft

When triggers are detected, the AI generates a 'Program Adjustment Draft' containing:

- Proposed changes: Specific modifications to upcoming sessions or goals
- Evidence citations: Links to the data that triggered the suggestion
- Rationale: Plain-language explanation of why this change is recommended
- Alternative options: 2-3 different approaches the coach could take
- Risk assessment: What happens if we don't make this change?

Coach Approval Workflow

Program Adjustment Drafts follow the same approval pattern as InsightCards:

48. Draft created → appears in Coach's 'Pending Adjustments' queue
49. Coach reviews evidence and proposed changes
50. Coach can: Approve as-is, Edit then approve, Reject with reason, or Request more context
51. Approved changes are applied to client's program
52. Client is notified with coach's message explaining the update
53. All actions logged in audit trail

Chat Integration

The chat interface can now explain program changes with full context:

- Client asks: 'Why did my program change?'
- Chat responds with: Current program state, what changed and when, the evidence that triggered the change (e.g., 'Your HRV dropped 15% this week'), and coach's explanation
- Grounded responses: Every explanation cites specific data points from the client's actual records

17D. Data Model Extensions for Programming

The following entities extend the data model from Section 4 to support programming:

Entity	Key Fields	Purpose
programs	id, coach_id, client_id, name, status, current_version	Top-level program container
program_versions	id, program_id, version_number, created_by, change_reason, created_at	Version history for programs
phases	id, program_version_id, name, focus, order, duration_weeks	Training blocks within program
microcycles	id, phase_id, week_number, notes	Weekly training structure
programmed_sessions	id, microcycle_id, day_of_week, session_type, name	Planned workout slots
programmed_exercises	id, session_id, exercise_id, sets, reps, weight, rest, notes, order	Exercise prescriptions
exercise_library	id, name, category, equipment, video_url, instructions	Master exercise database
session_completions	id, programmed_session_id, client_id, completed_at, status	Workout completion records
exercise_logs	id, session_completion_id, programmed_exercise_id, actual_sets, actual_reps, actual_weight, rpe, pain_flag, notes	Actual performance data
behavior_goals	id, coach_id, client_id, category, description, target, week_start, status	Weekly behavior targets
behavior_cues	id, goal_id, cue_text, reminder_time, active	Prompts supporting goals
behavior_checkins	id, goal_id, date, completed, friction_note, success_note	Daily goal tracking
program_adjustments	id, program_id, trigger_type, trigger_data, proposed_changes, status, coach_id, reviewed_at	AI-suggested modifications

17E. Programming Acceptance Tests

The following tests validate Programming Orchestration Layer functionality:

#	Test	Pass Criteria
17	Program creation	Coach creates multi-phase program with exercises; client sees schedule
18	Session completion logging	Client completes session; actual performance data persists correctly
19	Program versioning	Program edit creates new version; old version accessible; change reason logged
20	Behavior goal assignment	Coach assigns 1-3 weekly goals; client sees goals with cues
21	Behavior check-in	Client completes daily check-in; streak calculated correctly
22	AI adjustment trigger	New wearable data with declining HRV creates Program Adjustment Draft within 60 seconds
23	Adjustment approval flow	Coach approves adjustment; program updates; client notified with explanation
24	Chat explains changes	When asked 'Why did my program change?', chat cites specific evidence from client data
25	Escalation suggestion	Goal with <70% completion for 2 weeks triggers simplification draft for coach review

17F. Integration with Existing Systems

The Programming Orchestration Layer integrates with existing Xenios components:

Integration Point	How It Works
Session Recording (Section 2.1)	Transcripts feed into program adjustment triggers; session summaries can auto-populate exercise logs
Data Extraction (Section 2.4)	InBody, labs, and wearable data trigger program adjustments; extracted values inform exercise prescriptions
Coaching Analytics (Section 2.2)	Programming adherence data feeds into coaching performance metrics
Risk Detection (Section 2.3)	Poor program adherence contributes to client risk score
Insight Cards (Section 5-6)	Program Adjustment Drafts use same approval workflow as InsightCards
Chat Grounding (Test #4)	Chat can reference current program, upcoming sessions, and recent performance data

17G. Strategic Importance

The blunt truth: Without programming, Xenios risks becoming a very intelligent dashboard. Coaches do not pay for dashboards. They pay for outcomes delivered through a platform that helps them deliver a plan. Some coaches might pay for a dashboard and a platform that lets them program, but this would not differentiate Xenios in the market.

The Programming Orchestration Layer transforms Xenios from a 'data intelligence tool' into an 'outcome delivery system.' This is the critical difference between a product coaches use occasionally and a platform they cannot live without.

Key value propositions enabled by programming:

- Better coaches: Objective data on what programming approaches work for which client types
- More useful insights: Insights that directly inform program modifications, not just observations
- Coaching more clients: Systematic programming reduces per-client overhead
- Save time: AI-suggested adjustments reduce manual program management
- Viral spread: Clients who get results become ambassadors; programming enables measurable results

Document History

Version	Date	Author	Changes
1.0	Original	Xenios Team	Initial MVP PRD
2.0	January 2026	Samuel	Added Section 17: Programming Orchestration Layer including Fitness Programming, Behavior Programming, AI-Assisted Updates, Data Model Extensions, Acceptance Tests, and Integration Points